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1 C:\Users\thisa\AppData\Local\Microsoft\WindowsApps\
python3.13.exe C:\Users\thisa\Documents\
Software_Tech_1\Project\macro_project\src\console_app
.py
2
3 Macroinvertebrate Image Analysis System
4 =====
5 1. Show dataset summary (choose classes)
6 2. Generate EDA outputs
7 3. Train baseline classifier
8 4. Predict an image
9 5. Export dataset index to CSV
10 6. Show per-class image statistics
11 7. Exit
12 =====
13 Select an option: 1
14
15 --- Show Dataset Summary ---
16 Enter the path to your dataset folder: C:\Users\thisa
\Documents\Software_Tech_1\Project\macro_project\data
\raw\stream_macroinvertebrates
17
18 Found 17 classes:
19 1. Asellus sp
20 2. Baetidae sp
21 3. Elmia sp
22 4. Ephemerellidae
23 5. Erpobdella sp
24 6. Gammarus sp
25 7. Hydropsychidae sp
26 8. Leptophlebiidae sp
27 9. Leuctra sp
28 10. Limnias sp
29 11. Lymnaea sp
30 12. Nemoura sp
31 13. Oligochaeta sp
32 14. Sericostomatidae sp
33 15. Sialis sp
34 16. Simuliidae sp
35 17. Sphaerium sp
36
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37   Enter the numbers of at least 3 classes to include
    , separated by commas.
38   Example: 1,3,5
39   Your selection: 1, 2, 3
40
41   Selected classes: Asellus sp, Baetidae sp, Elmis sp
42
43   === Dataset Summary ===
44   Total Images: 448
45   Total Classes: 3
46   Mean Width: 600.0
47   Mean Height: 370.5
48   Count Asellus Sp: 294
49   Count Baetidae Sp: 109
50   Count Elmis Sp: 45
51
52   Macroinvertebrate Image Analysis System
53   =====
54   1. Show dataset summary (choose classes)
55   2. Generate EDA outputs
56   3. Train baseline classifier
57   4. Predict an image
58   5. Export dataset index to CSV
59   6. Show per-class image statistics
60   7. Exit
61   =====
62   Select an option: 2
63
64   --- Generate EDA Outputs ---
65   EDA outputs saved to outputs/eda/
66
67   Macroinvertebrate Image Analysis System
68   =====
69   1. Show dataset summary (choose classes)
70   2. Generate EDA outputs
71   3. Train baseline classifier
72   4. Predict an image
73   5. Export dataset index to CSV
74   6. Show per-class image statistics
75   7. Exit
76   =====
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77   Select an option: 3
78
79   --- Train Baseline Classifier ---
80   ✓ Confusion matrix saved to C:\Users\thisa\Documents
    \Software_Tech_1\Project\macro_project\outputs\
    models\confusion_matrix.png
81   ✓ Classification report saved to C:\Users\thisa\
    Documents\Software_Tech_1\Project\macro_project\
    outputs\models\classification_report.png
82   ✓ Model saved to C:\Users\thisa\Documents\
    Software_Tech_1\Project\macro_project\outputs\models
    \macro_classifier.joblib
83   Accuracy: 0.8556
84
85   Classification Report:
86           precision    recall  f1-score
87   support
88   Asellus sp          0.83      0.98      0.90
89   59
90   Baetidae sp         1.00      0.64      0.78
91   22
92   Elmis sp            0.83      0.56      0.67
93   9
94   accuracy
95   90
96   macro avg           0.89      0.72      0.78
97   90
98   weighted avg        0.87      0.86      0.85
99   90
100
101   Macroinvertebrate Image Analysis System
102   =====
103   1. Show dataset summary (choose classes)
104   2. Generate EDA outputs
105   3. Train baseline classifier
106   4. Predict an image
107   5. Export dataset index to CSV
108   6. Show per-class image statistics

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105     7. Exit
106     =====
107     Select an option: 4
108
109     --- Predict Image Class ---
110     Enter image path: wrong_file.jpg
111     Error: Could not read image: wrong_file.jpg
112 [ WARN:0@105.256] global loadsave.cpp:278 cv::
    findDecoder imread_('wrong_file.jpg'): can't open/
    read file: check file path/integrity
113
114     Macroinvertebrate Image Analysis System
115     =====
116     1. Show dataset summary (choose classes)
117     2. Generate EDA outputs
118     3. Train baseline classifier
119     4. Predict an image
120     5. Export dataset index to CSV
121     6. Show per-class image statistics
122     7. Exit
123     =====
124     Select an option:
```